

Abstract Algebra

: Lecture 19

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E
 $\uparrow$
 F roots.
 $f(x) \in F[x]$

Definition 1. Let E/F be a finite extension. Let $f(x) \in F[x]$. The smallest subfield of E which contains all of the roots of f is called the splitting field of f over F , or a splitting extension of F .

Example 2. Let $F = \mathbb{Q}$, and $f(x) = x^3 - 2 \in F[x]$. Then $\alpha = \sqrt[3]{2}$ is a root of f . Let $K = F(\alpha)$. Is K the splitting field of f over F ? $\rightarrow$ at most 3 roots.

No, since K does not contain the other roots of f . Let $\beta = \alpha e^{2\pi i/3}$, $\gamma = \alpha e^{4\pi i/3}$ then β, γ are also roots of f . Let $\omega = e^{2\pi i/3}$, the splitting field of f over F should be $F(\alpha, \omega) = F(\alpha + c\omega)$ for some $c \in \mathbb{Q} - \{0\}$.

Definition 3. For E/F an automorphism σ of E which fixes F pointwise is called an F -automorphism of E . All automorphism of E which fix F pointwise form a group, called the Galois group of E/F , denoted by $\text{Gal}(E/F)$ or $\text{Gal}(E : F)$. If E is the splitting field of some $f(x) \in F[x]$, then $\text{Gal}(E/F)$ is called the Galois group of f over F , denoted by $\text{Gal}(f)$.

Proposition 4. Let $F < K \leq E$, K is the splitting field of some $f(x) \in F[x]$ over F .

(1). K is unique; $\checkmark$.

(2). Each F -automorphism of E induces an F -automorphism of K . $\checkmark$ Which is due to the fact that σ fixes $f(x)$ and permutes the roots of $f(x)$.

Example 5. Let $F = \mathbb{R}$ and $E = \mathbb{C}$. Then E is a splitting field of F and $\text{Gal}(E/F)$ is isomorphic to $Z_2 = \langle \sigma \rangle$ where $\sigma : a + bi \mapsto a - bi$, $a, b \in \mathbb{R}$. Actually, $E \simeq F[x]/(x^2 + 1)$. $\rightarrow \text{Gal}(\mathbb{C}/\mathbb{R}) \cong \langle \sigma \rangle$

Example 6. Let $E = \mathbb{Q}(\sqrt{2}) \simeq \mathbb{Q}[x]/(x^2 - 2)$. Then what is the Galois group of $E/\mathbb{Q}$? $\sigma : x \mapsto \bar{x}$.
 $Z_2 = \langle \sigma \rangle$. Where $\sigma : a + b\sqrt{2} \mapsto a - b\sqrt{2}$, $a, b \in \mathbb{Q}$ since σ fixes $x^2 - 2$. $\rightarrow \text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \cong \langle \sigma \rangle$

Example 7. Let $x^3 - 2 \in \mathbb{Q}[x]$. Then the splitting field of $x^3 - 2$ over $\mathbb{Q}$ is $E = \mathbb{Q}(\alpha, \omega)$. $\rightarrow$ 6 needs to permute them. $\sigma : a + b\sqrt{2} \mapsto a - b\sqrt{2}$.
Then root of $x^3 - 2$ are $\alpha, \omega\alpha, \omega^2\alpha$. Let $\rho : \alpha \mapsto \omega\alpha \mapsto \omega^2\alpha$, $\sigma : \alpha \mapsto \alpha$, $(\omega\alpha, \omega^2\alpha) \mapsto (\omega^2\alpha, \omega\alpha)$. Then $\text{Gal}(E/\mathbb{Q}) = \langle \rho, \sigma \rangle \simeq S_3$.
 $\alpha \mapsto \omega\alpha$
 $\omega\alpha \mapsto \omega^2\alpha$
 $\omega^2\alpha \mapsto \alpha$

Example 8. $E = \mathbb{Q}(\alpha)$ is not the splitting field of $x^3 - 2$ over $\mathbb{Q}$. Let $F = \mathbb{Q}(\omega)$. Is $K = F(\alpha)$ the splitting field of $x^3 - 2$ over F ? Yes. $\checkmark$.
 $e^{2\pi i/3}$.

$\text{Gal}(E/\mathbb{Q}) = ?$ Suppose $\sigma \in \text{Gal}(E/\mathbb{Q})$ then let $(2^{1/3})^\sigma = \beta$, we have $2 = 2^\sigma = ((2^{1/3})^\sigma)^\sigma = \beta^3$ i.e. $\beta = 2^{1/3}$ i.e. $\sigma = 1$. Hence $\text{Gal}(E/\mathbb{Q}) = 1$. $\checkmark$.

$\alpha \mapsto \alpha$.

$\text{Aut}(\mathbb{Q}) = \{\text{Id}\}$.

$\rightarrow$ Recall: $\text{Aut}(\mathbb{R}) = \{\text{id}\}$ as field automorphism.

$$\mathbb{Q}(w, \alpha) / \mathbb{Q}(\alpha).$$

$$\alpha \mapsto \alpha.$$

$$\mathbb{Q}(w, \alpha)$$

$$(\mathbb{Q}(w))(\alpha) / (\mathbb{Q}(w)).$$

$$\alpha \mapsto ?$$

$$\text{rd.}$$

$$\alpha \mapsto w\alpha \mapsto w^2\alpha \mapsto \alpha.$$

$$\alpha \mapsto w^2\alpha \mapsto w\alpha \mapsto \alpha.$$

①②③ forms Z_3

6 is impossible.

Exercise 9. What is $\text{Gal}(K/F)$? It's Z_3 .

Example 10. Let $L = \mathbb{Q}(\alpha)$, $\text{Gal}(E/L) = ?$ It's Z_2 .

Theorem 11. Let L be a splitting field of $g(x) \in K[x]$. Then for any irreducible polynomial $f(x)$, whenever $f(x)$ has a root in L , then $f(x)$ splits in L .

证明. Let $L = K(\alpha_1, \dots, \alpha_n)$, let $f(x) \in K[x]$ be irreducible. Let α, β be two roots of $f(x)$ in L s.t. $\alpha \in L$. We aim to prove $\beta \in L$. $\star, \checkmark. \iff [L(\beta) : L] = 1$.

$L(\alpha) = K(\alpha_1, \dots, \alpha_n)(\alpha) = K(\alpha)(\alpha_1, \dots, \alpha_n)$, $L(\beta) = K(\alpha_1, \dots, \alpha_n)(\beta) = K(\beta)(\alpha_1, \dots, \alpha_n)$. $L(\beta)$ is the splitting field of g on $K(\beta)$ and $L(\alpha)$ is the splitting field of g on $K(\alpha)$. Notice that $[L(\alpha) : K(\alpha)] = [L(\beta) : K(\beta)]$ and $[K(\alpha) : K] = [K(\beta) : K]$, as α, β are two roots of the irreducible polynomial $f(x) \in K[x]$. $[L(\beta) : L][L : K] = [L(\beta) : K] = [L(\beta) : K(\beta)][K(\beta) : K] = [L(\alpha) : K(\alpha)][K(\alpha) : K] = [L(\alpha) : K] = [L(\alpha) : L][L : K] = [L : K]$. i.e. $[L(\beta) : L] = 1$. $\checkmark$.

Theorem 12. Let $F < L < E$. Then L is splitting extension of F iff $\text{Gal}(E/L) \triangleleft \text{Gal}(E/F)$.

证明. next time $\square$

Definition 13. A splitting extension is called a normal extension. *splitting extension $\iff$ normal extension.*

Example 14. Let $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$, $g(x) = (x^2 - 2)(x^2 - 3)$, $f(x) = x^4 - 10x^2 + 1$.

(1). Different polynomials may have the same splitting field.

(2). $\text{Gal}(g) = \langle \sigma \rangle \times \langle \tau \rangle \simeq Z_2 \times Z_2$, where $\sigma : \sqrt{2} \mapsto -\sqrt{2}$ and $\tau : \sqrt{3} \mapsto -\sqrt{3}$. $\text{Gal}(f) = Z_2 \times Z_2$, they are equal. i.e. Different polynomials may have the same Galois group. But this group acts on the roots of g is not transitive, and acts on the roots of f is transitive, since f is irreducible.

permutes $\sqrt{2}, -\sqrt{2}$ permutes $\sqrt{3}, -\sqrt{3}$.